



PHYSICS HSSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which of the following equations is dimensionally correct?	$v_f = v_i + at$	$v_f^2 = v_i^2 + 2at$	$s = v_i t + \frac{1}{2}at$	$F = ma^2$	☐	☐	☐	☐
2. Pascal (Pa) is equal to:	N/m	N/m ²	N ² /m ²	kg/m	☐	☐	☐	☐
3. Time of flight of a projectile depends on:	Horizontal velocity	Mass	Vertical component of velocity	Acceleration only	☐	☐	☐	☐
4. A car accelerates uniformly from rest to 60ms ⁻¹ in 5 seconds. What distance does it cover during this time?	150m	450m	300m	600m	☐	☐	☐	☐
5. Which of the following is a correct way to represent Young's Modulus?	$\frac{F/A}{\Delta L/L}$	$\frac{F.A}{\Delta L}$	$\frac{F}{A.\Delta L}$	$\frac{F.\Delta L}{L.A}$	☐	☐	☐	☐
6. In a steady, incompressible flow, if the cross-sectional area of a pipe decreases to one-third of its original size, what happens to the fluid velocity?	It decreases to one-third	It remains the same	It triples	It becomes nine times larger	☐	☐	☐	☐
7. Direction of centripetal force is always:	Tangential	Perpendicular	Radially inward	Outward	☐	☐	☐	☐
8. If the temperature of heat source is doubled than before, efficiency increases by:	1.5 times	2 times	2.5 times	3 times	☐	☐	☐	☐
9. Antinode in a wave is point of:	Zero displacement	Maximum displacement	Zero velocity	No wave	☐	☐	☐	☐
1. What is the motion of an electron moving at an angle of 45° to a uniform magnetic field?	Linear motion	Circular motion	Parabolic motion	Helical motion	☐	☐	☐	☐



	Question	A	B	C	D	A	B	C	D
11.	A potentiometer is used to compare:	Resistance	Current	Potential difference	Power	☐	☐	☐	☐
12.	Electric field strength is:	Work done	Energy	Force per unit charge	Charge per force	☐	☐	☐	☐
13.	Postulates of special theory of relativity applies to:	Only light	Only motion	All physical laws	Only energy	☐	☐	☐	☐
14.	Correct Mass-energy relation is:	$E = mv$	$E = mc^2$	$E = mg$	$E = gh$	☐	☐	☐	☐
15.	Proton is made up of:	2 down and 1 up quark	1 down and 2 up quarks	2 down and 2 up quarks	4 down and 2 up quarks	☐	☐	☐	☐
16.	If distance from axis increases, moment of inertia:	Decreases	Increases	Remains constant	Becomes zero	☐	☐	☐	☐
17.	Correct relation between torque and angular acceleration is:	$\tau = \frac{IR}{\alpha}$	$\tau = \frac{\alpha}{I}$	$\tau = \frac{I}{2}\alpha$	$\tau = m\alpha$	☐	☐	☐	☐

—1HA-I 26004-(30041) (B) —

- $S = v_i t + \frac{1}{2}at^2$
- $I = \frac{1}{2}MR^2$
- $F_c = \frac{mv^2}{r}$
- $E = \frac{F}{q}$
- $V = \frac{W}{q}$
- $\eta = 1 - \frac{T_2}{T_1}$
- $I = \frac{2}{5}MR^2$
- $I = mr^2$
- $A_1v_1 = A_2v_2$
- $v_f = v_i + at$
- $E = mc^2$
- $Av = \text{constant}$

ROLL NUMBER						





PHYSICS HSSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

28

Time allowed: 2:35 Hours

Total Marks Sections B and C: 63

SECTION – B (Marks 39)

Q. 2 Answer the following parts briefly.

(13 x 3 = 39)

(i)	Deduce the dimensions of k from Coulomb's law $\left(F = k \frac{q_1 q_2}{r^2}\right)$.	03	OR	Describe any three properties of cross product of two vectors.	1x3
(ii)	Describe photon as a force carrier.	03	OR	Distinguish between the terms elastic deformation and plastic deformation.	03
(iii)	Show that: $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$	03	OR	Show that ε and $\frac{\Delta \phi}{\Delta t}$ have the same units.	03
(iv)	Differentiate between conservative and non-conservative forces with examples.	03	OR	Distinguish between turbulent and streamline flow.	03
(v)	Calculate the resistance of wire 10 m long that has diameter of 3mm and resistivity of $2.63 \times 10^{-2} \Omega m$.	03	OR	A body of mass $(27.2 \pm 0.1) kg$ has a volume of $(2.0 \pm 0.1) m^3$. Calculate the density of the body? Find uncertainty in the density measurement.	03
(vi)	Derive the relationship $P.E = \frac{1}{2} \left[\frac{YA}{L} \right] x^2$	03	OR	In thermodynamics briefly describe: Reversible process Irreversible process Cyclic process	1x3
(vii)	A 5 kg shell is fired with a muzzle velocity of $700 ms^{-1}$ from the gun barrel 2.4m long. Calculate the average value of: The acceleration The time to come out of the barrel	03	OR	Describe the working of Atomizers in perfume bottles.	03
(viii)	What is meant by precision? Also write the importance of taking multiple readings in an experiment.	1+2	OR	Find the area of the parallelogram whose adjacent sides are given by the vectors: $A = (\hat{i} + 6\hat{j} + 2\hat{k})m$ and $B = (7\hat{i} + \hat{j} + 5\hat{k})m$.	03
(ix)	Briefly explain light dependent resistor (LDR). Give one application of LDR.	2+1	OR	The front wheel of a tractor travels 700 revolutions while the rear wheel 280 in a time interval of 40 seconds. Find their angular velocities.	03
(x)	Compare constructive and destructive interference.	1.5 + 1.5	OR	Derive a relationship between torque and angular acceleration.	03
(xi)	State first law of thermodynamics. Briefly explain the term internal energy.	1.5 + 1.5	OR	What is meant by potential gradient? Derive its formula.	03
(xii)	Briefly explain the relationship between pressure and volume of a gas at constant temperature. Show it graphically as well.	2+1	OR	What is meant by superfluid? How is superfluid made?	1.5 + 1.5
(xiii)	Water flows through a 1cm diameter pipe with a speed of $1 ms^{-1}$. What should be the diameter of the nozzle if the water is to emerge at $2.1 ms^{-1}$?	03	OR	A stationary observer hears the sound of a car horn at a frequency that is 15% higher than the horn's actual frequency. If the speed of sound is $343 ms^{-1}$, what is the speed of the car as it approaches the observer?	03

SECTION – C (Marks 24)

Attempt the following questions.

Q.3	Derive expressions for: a. Range of projectile b. Height of projectile	3+3	OR	a. State postulates of special theory of relativity. b. Briefly discuss time dilation and length contraction. c. At what speed would the mass of the proton be tripled? The rest mass of a proton is $1.673 \times 10^{-27} kg$.	2+2 +2
Q.4	a. State Bernoulli's equation. b. Derive velocity of efflux using Torricelli's theorem. Draw diagram as well.	1+3 +2	OR	Derive an expression for a magnetic force experienced by a moving charge in a uniform magnetic field. Also discuss the direction of that force.	4+2
Q.5	a. Discuss working of a refrigerator. b. How increase in Entropy means degradation of energy? Explain with an example.	3+3	OR	a. State Kirchoff's laws for current and voltage. b. There is a copper coil having 2000 turns with $0.8 mm^2$ cross-section area, the length per turn is 80cm. Calculate the resistance of the coil.	3+3
Q.6	a. Derive the expression for Kinetic energy. b. State work energy theorem in a resistive medium.	3+3	OR	Derive f_1, f_2 and f_3 for modes of vibration in a closed organ pipe.	2+2 +2

1HA-I 26004 (B)

$$F = G \frac{m_1 m_2}{r^2}$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$KE = \frac{1}{2} mv^2$$

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$v = \sqrt{2gh}$$

$$R = \frac{\rho L}{A}$$

$$PE = mgh$$

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

$$I = I_o \cos^2 \theta$$

$$A = \pi r^2$$

$$P = \vec{F} \cdot \vec{v}$$

$$A_1 v_1 = A_2 v_2$$

$$c = 3 \times 10^8 ms^{-1}$$

$$\text{Work} = \vec{F} \cdot \vec{x}$$

$$MV = mv$$

$$v_f = v_i + at$$

$$2aS = v_f^2 - v_i^2$$

$$v = r\omega$$

$$\omega = \frac{\theta}{t}$$

$$N = \frac{S}{2\pi r}$$

$$m = \frac{m_o}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$f_o = f_s \frac{v}{v \pm v_s}$$

$$H = \frac{v_i^2 \sin^2 \theta}{2g}$$

$$E = -\frac{\Delta V}{\Delta r}$$

$$\theta = 2\pi r$$

$$\rho_{\text{copper}} = 0.02 \mu \Omega m$$



PHYSICS HSSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which equation correctly relates torque (τ), moment of inertia (I), and angular acceleration (α)?	$\tau = I^2 \alpha$	$\tau = \frac{I}{\alpha}$	$\tau = I + \alpha$	$\tau = I \alpha$	☐	☐	☐	☐
2. The dimensional formula of velocity is:	$[M^0 L^0 T^0]$	$[LT^{-1}]$	$[LT^{-2}]$	$[L^2 T^{-1}]$	☐	☐	☐	☐
3. Joule can be expressed as:	$kg m^{-2} s^{-2}$	$kg m s^{-1}$	$kg m^2 s^{-1}$	$kg m^2 s^{-2}$	☐	☐	☐	☐
4. Horizontal velocity in projectile motion:	is zero at max height	increases	decreases	remains constant	☐	☐	☐	☐
5. A bus travelling at $10ms^{-1}$ accelerates uniformly at $2ms^{-2}$. Calculate the velocity of the bus after 5 s.	$0.02ms^{-1}$	$20ms^{-1}$	$2.0ms^{-1}$	$0.2ms^{-1}$	☐	☐	☐	☐
6. Which of the following is a correct way to represent Young's Modulus?	$\frac{F \Delta L}{L A}$	$\frac{F / A}{\Delta L / L}$	$\frac{F A}{\Delta L}$	$\frac{F}{A \Delta L}$	☐	☐	☐	☐
7. Continuity equation is based on:	Force	Energy conservation	Momentum	Mass conservation	☐	☐	☐	☐
8. If velocity of a body moving in circular path doubles, centripetal force becomes:	Four times	Same	Double	Half	☐	☐	☐	☐
9. According to the first law of thermodynamics, if 200 Joules of heat is added to a system and 80 Joules of work is done by the system, the internal energy change is:	120J decrease	80J increase	120 J increase	280J increase	☐	☐	☐	☐
10. Stationary waves are formed by:	Diffraction	Reflection only	Superposition of waves	Refraction	☐	☐	☐	☐



	Question	A	B	C	D	A	B	C	D
11.	Path of charged particle in magnetic field is always:	Rondon	Straight	Parabolic	Circular	☐	☐	☐	☐
12.	At balance point, current in galvanometer is:	Zero	Maximum	Minimum	Infinite	☐	☐	☐	☐
13.	If 800J energy is used to move a charge of 4C between two points in an electric field. Potential difference is:	5V	100V	200V	$5 \times 10^{-3} V$	☐	☐	☐	☐
14.	The rest energy of mass 2.00g is:	$1.80 \times 10^{18} J$	$180 \times 10^{12} J$	$1.80 \times 10^{14} J$	$1.80 \times 10^{16} J$	☐	☐	☐	☐
15.	According to special theory of relativity time dilation means:	Time increases for moving observer	Time decreases for moving observer	Time becomes zero	Time is constant for moving observer	☐	☐	☐	☐
16.	Neutrino is:	Magnetic particle	Charged particle	Heavy particle	Neutral particle	☐	☐	☐	☐
17.	A solid cylinder with mass $M=5\text{kg}$ and radius $R=0.3\text{m}$ rotates about its central axis. Calculate the moment of inertia of the cylinder.	0.23kgm^2	5.3kgm^2	4.7kgm^2	1.5kgm^2	☐	☐	☐	☐

—1HA-I 26004-32042 (D) —

- $v_f = v_i + at$
- $I = \frac{1}{2}MR^2$
- $F_c = \frac{mv^2}{r}$
- $E = \frac{F}{q}$
- $V = \frac{W}{q}$
- $E = mc^2$
- $I = \frac{2}{5}MR^2$
- $I = MR^2$

ROLL NUMBER						





PHYSICS HSSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

30

Time allowed: 2:35 Hours

Total Marks Sections B and C: 63

SECTION – B (Marks 39)

Q. 2 Answer the following parts briefly.

(13 x 3 = 39)

(i)	Deduce the dimensions of universal gravitational constant (G). $\left(F = G \frac{m_1 m_2}{r^2}\right)$	03	OR	Describe any three properties of dot product of two vectors.	1x3
(ii)	Give Comparison of amorphous and polymeric solids. (Three points)	1x3	OR	State the Higgs Boson as a fundamental particle which is responsible for the particle's mass.	03
(iii)	The magnitudes of dot and cross product of two vectors are $6\sqrt{3}$ and 6 respectively. Find the angle between the vectors.	03	OR	How a seismometer uses electromagnetic induction to detect earthquakes? Explain briefly.	03
(iv)	In a science experiment a 3.00 kg water rocket, is launched from ground. The rocket's total energy at the top of its flight is 2352 J. a. What was the rocket's launching speed? b. What height did the rocket went?	03	OR	Find the power delivered by the engine for attaining velocity $(3,4) \text{ ms}^{-1}$ while it exerts a force $(8, -2) \text{ N}$.	03
(v)	Calculate the resistance of a wire 10 m long that has diameter of 2 mm and resistivity of $2.63 \times 10^{-2} \Omega \text{m}$.	03	OR	Differentiate between accuracy and precision of measurements. (Give a graphical example)	03
(vi)	Briefly explain brittle and ductile materials. (In context of elastic and plastic deformation)	1.5 +1.5	OR	State first law of thermodynamics. Discuss its significance.	1+2
(vii)	Evaluate effect of air resistance on projectile motion. Show graphically.	2+1	OR	Briefly explain Magnus effect (an application of Bernoulli Effect). Draw diagram as well.	2+1
(viii)	Write any three limitations of dimensional analysis.	03	OR	Briefly discuss any three properties of an ideal fluid.	1x3
(ix)	What is magnetic flux? Also describe magnetic flux density and write its formula.	1+2	OR	Write any three characteristics of stationary waves.	1x3
(x)	What are beats? How are they produced?	1+2	OR	Determine the equivalent value of 1 radian in degrees.	03
(xi)	Briefly describe: a. Unit cell and its types b. Crystal lattice	2+1	OR	Derive the relation between Torque and angular acceleration.	03
(xii)	What is a light dependent resistor (LDR)? Why is LDR useful in automatic circuits?	1+2	OR	Explain the use of beats in tuning musical instruments.	03
(xiii)	The orbit of an electron revolving around the nucleus of an atoms is considered to be circular of radius 50 pm and the period of motion is 0.2 fs. Find out angular velocity of electron (ω).	03	OR	The electric field at a point due to a point charge is 26 N/C and the electric potential at that point is 13 J/C. Calculate the distance of the point from the charge.	03

SECTION – C (Marks 24)

Attempt the following questions.

Q.3	a. What is meant by polarization. b. Discuss one application of polarization. c. A polarized light of intensity I_0 is passed through another polarizer whose axis makes an angle of 60° with the axis of the polaroid filter. What is the intensity of transmitted polarized light from second polaroid filter?	1+2 +3	OR	a. State Archimedes principle. b. A water storage tank in a residential area develops a small leak near its lower end. The top of the water in the tank is 10m above the point of the leak. (i) With what speed does the water rush from the hole? (ii) If the hole has an area of 0.080 cm^2 , how much water flows out in one second?	1+5
Q.4	Describe elastic collision in one dimension. Derive relation to show the relative velocities before and after collision.	1+5	OR	Briefly explain Elastic Potential Energy. Derive its mathematical formula.	2+4
Q.5	State working principle of Carnot's Engine. With the help of graph discuss its all four processes in detail.	1+1 +4	OR	A ball of 100gm is thrown vertically upward at a speed of 25m/s. If no energy is lost, determine the height it would reach. If the ball only rises to 25m, calculate the work done against the air resistance. Also calculate the force of friction.	06
Q.6	a. What is meant by term protentional gradient? Derive its formula. b. Show that $\frac{\text{volt}}{\text{meter}} = \frac{\text{newton}}{\text{columb}}$	3+3	OR	a. State postulates of special theory of relativity. b. Briefly discuss length contraction as a consequence of special relativity. c. The length of spaceship is 100m long. What will be its length if the spaceship moves with a speed of $0.99c$?	2+1 +3

— 1HA-I 26004 (D) —

$$F = G \frac{m_1 m_2}{r^2}$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$KE = \frac{1}{2} mv^2$$

$$\text{Work} = \vec{F} \cdot \vec{x}$$

$$v = \sqrt{2gh}$$

$$R = \frac{\rho L}{A}$$

$$PE = mgh$$

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

$$I = I_0 \cos^2 \theta$$

$$A = \pi r^2$$

$$P = \vec{F} \cdot \vec{v}$$

$$A_1 v_1 = A_2 v_2$$

$$c = 3 \times 10^8 \text{ ms}^{-1}$$

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$MV = mv$$

$$v_f = v_i + at$$

$$2aS = v_f^2 - v_i^2$$

$$v = r\omega$$

$$\omega = \frac{\theta}{t}$$

$$N = \frac{S}{2\pi r}$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$f_o = f_s \frac{v}{v \pm v_s}$$

$$H = \frac{v_i^2 \sin^2 \theta}{2g}$$

$$E = -\frac{\Delta V}{\Delta r}$$

$$\theta = 2\pi r$$

$$\rho_{\text{copper}} = 0.02 \mu\Omega \text{m}$$